Course 02323/02402 Introduction to Statistics

Lecture 10: Inference for proportions

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Agenda

1. Introduction
2. Random variable for proportion
3. Confidence interval for a single proportion
   - Sample size and experiment planning
4. Hypothesis test for a single proportion
5. Confidence Interval and Hypothesis Test for Two Proportions
6. Hypothesis test for multiple proportions
7. Statistics for contingency tables
8. Summary

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Different analyses and data types

Means in quantitative data

- Hypothesis test for a single mean based on one sample
- Hypothesis test for two means based on two samples
- Hypothesis test for multiple means based on several samples (ANOVA - coming later).

Different analyses and data types

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Today: Proportions in qualitative data

- Hypothesis test for a single proportion based on one sample.
- Hypothesis test for two proportions based on two samples.
- Hypothesis test for multiple proportions based on several samples.

Example of proportions

Left-handed:

The proportion of left-handed individuals
(in Denmark vs. Sweden)

or:

Female engineering students:

The proportion of female engineering students
(in different fields of study)

Kahoot!
(x4)

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Estimation of proportions

- We define the random variable P as the number of "successes" (X) out of a total (n):

$$P = \frac{X}{n}$$

- From sample data with x "successes" (sample size n), we estimate the proportion as:

$$\hat{p} = \frac{x}{n}$$

Note:

- $P \in [0; 1]$.
- p is the "true" population probability of a "success".

Binomial distribution

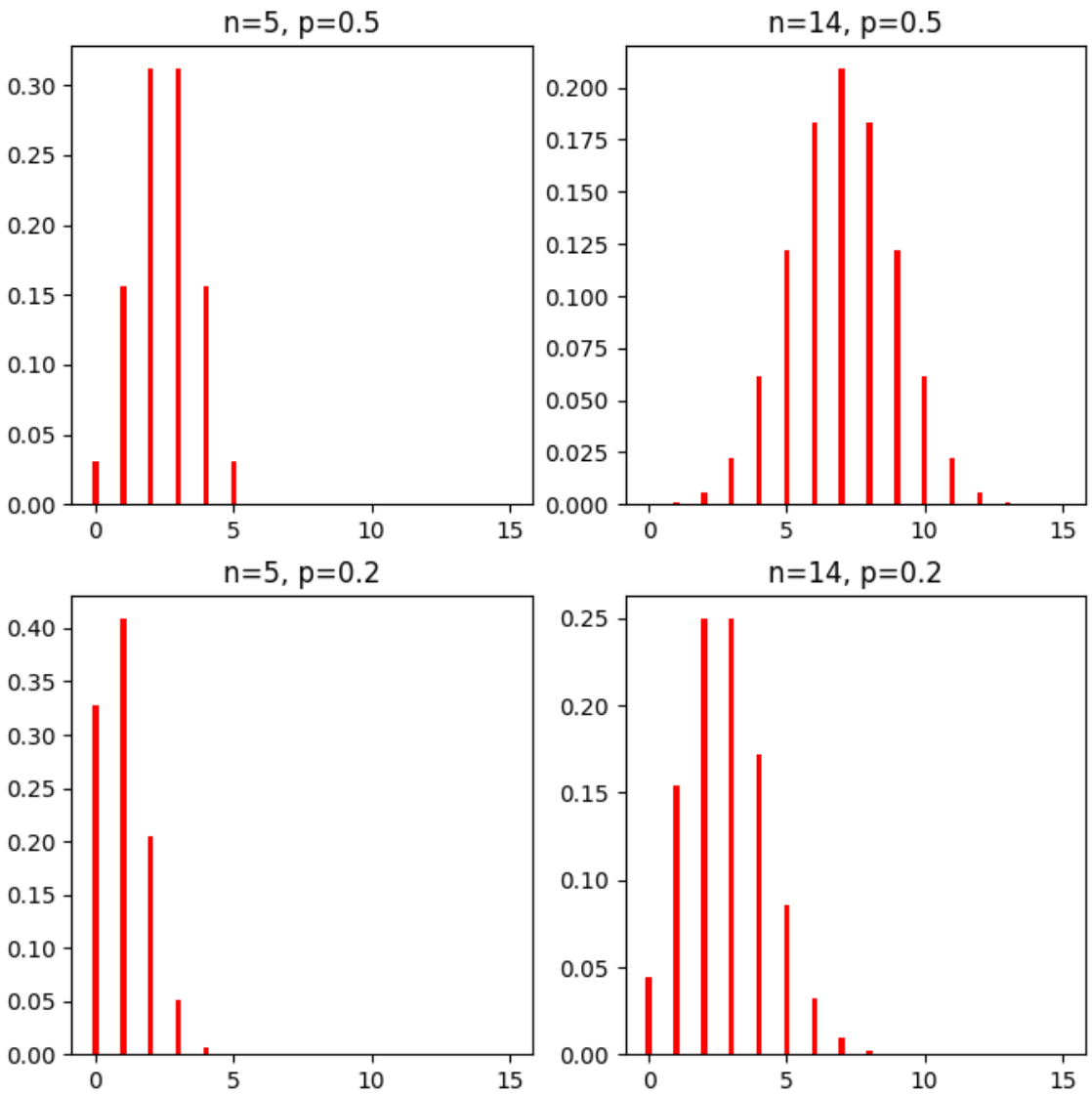
The number of "successes" (X) follows a binomial distribution with the density function:

$$f(x;n,p) = \binom{n}{x} p^x (1-p)^{n-x}$$

Mean and variance in the binomial distribution, section 2.21

$$\begin{aligned} \mathbf{E}[X] &= np \\ \mathbf{V}[X] &= np(1-p) \end{aligned}$$

Examples of binomial distributions (from week 2)



Mean and variance for proportions

Mean and variance for the proportion P:

$$\mathbf{E}[P] = \mathbf{E}\left[\frac{X}{n}\right] = \frac{np}{n} = p$$

$$\mathbf{V}[P] = \mathbf{V}\left[\frac{X}{n}\right] = \frac{1}{n^2}\mathbf{V}[X] = \frac{p(1-p)}{n}$$

Thus, we can define:

$$\sigma_P = \sqrt{\frac{p(1-p)}{n}}$$

Note:

σ_P is largest when $p = 1/2$.

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Confidence interval for a single proportion

Method 7.3

If the sample is large, then the $(1-\alpha)$-confidence interval for p is given by:

$$\hat{p} \pm z_{1-\alpha/2}\, \sigma_{\hat{p}}$$

In practice, $\hat{p}$ is substituted for p in the formula $\sigma_{\hat{p}=\sqrt{\hat{p}(1-\hat{p})/n}}$

Confidence interval for a single proportion

Method 7.3

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How?

This follows from approximating the binomial distribution with the normal distribution.

Confidence interval for a single proportion

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How?

This follows from approximating the binomial distribution with the normal distribution.

Rule of thumb

Assume $X \sim \text{bin}(n, p)$. The normal distribution is a good approximation for the binomial distribution if np and $n(1-p)$ (expected number of successes and failures) are both at least 15.

Example in Python

- Go to today's Python notebook in VS Code
  - "Example: Normal approximation of binomial distribution"



Confidence interval for a single proportion in small samples

There exist several ways of expressing a valid confidence interval for p in small sample cases, that is, when either $np \leq 15$ or $n(1-p) \leq 15$. Book mentions three of these – only for the last one, we give the explicit formula:

"Plus 2" approach (Remark 7.7)

If the sample is not large, use $\tilde{p} = \frac{x+2}{n+4}$, $\tilde{n} = n+4$
In the confidence interval, insert:

$$\tilde{p} \pm z_{1-\alpha/2} \sqrt{\tilde{p}(1-\tilde{p})/\tilde{n}}$$

Example in Python

In Python, "exact interval", for all n and p:

Python Code

```
test = stats.binomtest(x, n)
ci = test.proportion_ci(confidence_level=0.95)
```

Assuming Normality, not using continuity correction (making the best approximation of discrete probabilities):

Python Code

```
z_obs, p_value = smprop.proportions_ztest
(count=518, nobs=1154, value= 0.5,prop_var =0.5)
ci_low, ci_high = smprop.proportion_confint(
count=518,nobs=1154, alpha=0.05,method= "normal")
```

- Go to today's Python notebook in VS Code
  - "Example: Confidence interval of proportion for left-handed"



Margin of error (ME)

Margin of error

at a $(1-\alpha)$-confidence level is:

$$ME = z_{1-\alpha/2}\sqrt{\frac{p(1-p)}{n}}$$

where we estimate p with $\hat{p} = \frac{x}{n}$.

Margin of error:

- Corresponds to half the width of the $(1-\alpha)$-confidence interval.
- Describes the expected precision (minimum desired precision) of the estimate $\hat{p}$.

Precision and sample size

Experiment planning:

How large does the sample size need to be to achieve a given precision?

Method 7.13

If you want an expected (given) margin of error (ME) in a $(1-\alpha)$-confidence interval, the required sample size is:

$$n = p(1-p)\left(\frac{z_{1-\alpha/2}}{\text{ME}}\right)^2,$$

where p is a reasonable guess.

Precision and sample size

Method 7.13

If you want an expected (given) margin of error (ME) in a $(1-\alpha)$-confidence interval but do *not* have a reasonable guess of p, the required sample size is:

$$n = \frac{1}{4}\left(\frac{z_{1-\alpha/2}}{\text{ME}}\right)^2,$$

because the "worst case" is $p = \frac{1}{2}$.

Example 1 – continued

Left-handed:

Assume we want ME $= 0.01$ (where $\alpha = 0.05$) – what should n be?

Example 1 – continued

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Based on the sample, we guess $p = 0.10$:

$$n = 0.1 \cdot 0.9 \left(\frac{1.96}{0.01} \right)^2 = 3467.4 \approx 3468$$

Example 1 – continued

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Assume we want ME $= 0.01$ (where $\alpha = 0.05$) – what should n be?

Based on the sample, we guess $p = 0.10$:

$$n = 0.1 \cdot 0.9 \left(\frac{1.96}{0.01} \right)^2 = 3467.4 \approx 3468$$

Without assuming a guess for p:

$$n = \frac{1}{4} \left(\frac{1.96}{0.01} \right)^2 = 9604$$

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Steps in a hypothesis test – Overview (review!)

1. Formulate the null hypothesis and choose a significance level α.
2. Calculate the observed test statistic.
3. Calculate the p-value from the observed test statistic and the relevant distribution.
4. Compare the p-value with the significance level α and conclude.

Alternatively: Compare the observed test statistic with critical values and conclude.

Hypothesis test for a single proportion

We consider a null and alternative hypothesis for a single proportion p and choose a significance level α:

$$H_0: \quad p = p_0,$$
$$H_1: \quad p \neq p_0.$$

As usual, reject H_0 or accept H_0.

Hypothesis test: Test statistic

Theorem 7.10 and Method 7.11

If the sample is large enough ($np_0 > 15$ and $n(1-p_0) > 15$), we use the test statistic:

$$z_{\text{obs}} = \frac{x - np_0}{\sqrt{np_0(1-p_0)}} = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$$

Under the null hypothesis, the test statistic approximately follows a standard normal distribution.

Hypothesis test: p-value and conclusion (Method 7.11)

Find the p-value (evidence against the null hypothesis):

- $2P(Z > |z_{\text{obs}}|)$

Testing using critical values:

We reject the null hypothesis if $z_{\text{obs}} < -z_{1-\alpha/2}$ or $z_{\text{obs}} > z_{1-\alpha/2}$.

Example in Python

- Go to today's Python notebook in VS Code
  - "Example: Hypothesis test for proportion of left-handed individuals"

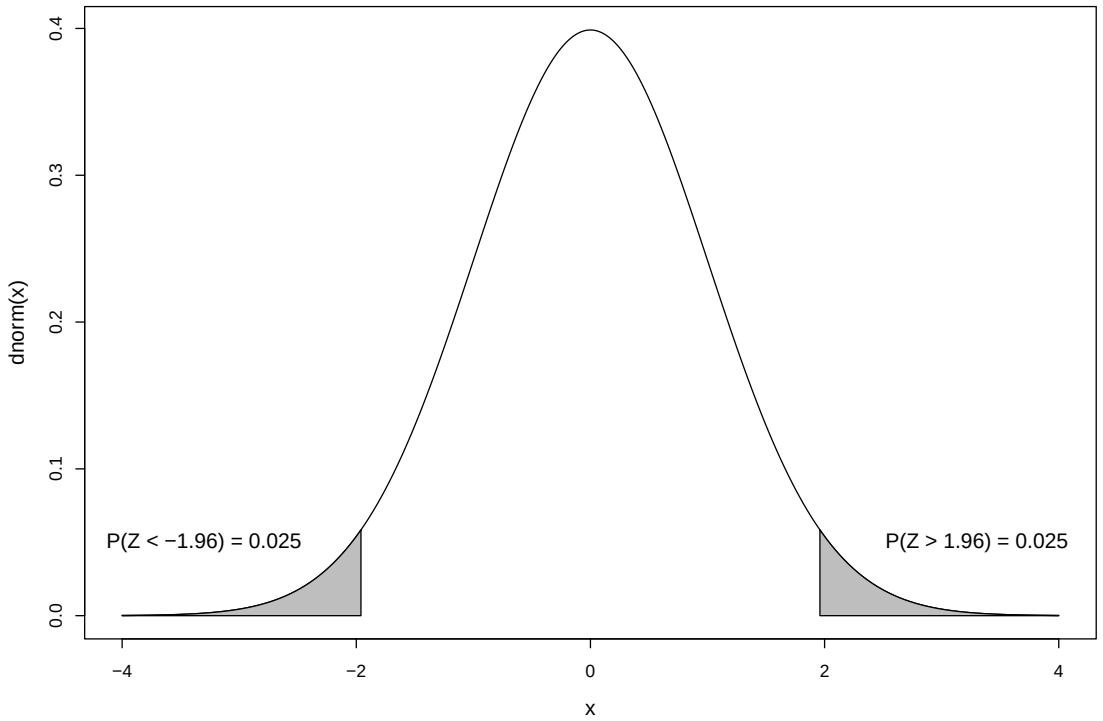


Example 1 – Continued

Using critical values:

$$z_{0.975} = 1.96$$

Since $z_{\text{obs}} = -8$ is (much) less than -1.96, we reject the null hypothesis.



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Confidence Interval for the Difference of Two Proportions

Method 7.15

$$(\hat{p}_1 - \hat{p}_2) \pm z_{1-\alpha/2} \cdot \hat{\sigma}_{\hat{p}_1 - \hat{p}_2}$$

where

$$\hat{\sigma}_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

Rule of Thumb

Both $n_i \hat{p}_i \geq 10$ and $n_i(1-\hat{p}_i) \geq 10$ for $i = 1, 2$.

Hypothesis Test for the Difference of Two Proportions - Method 7.18

Hypothesis Test for Two Proportions

When comparing two proportions (shown here for a two-sided alternative hypothesis):

$$\begin{aligned} H_0: &\quad p_1 = p_2, \\ H_1: &\quad p_1 \neq p_2. \end{aligned}$$

Use the test statistic

$$z_{\text{obs}} = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}, \quad \text{where} \quad \hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

Example 2

Is there a relationship between birth control pill use and the risk of heart clots?

A study (USA, 1975) investigated the association between birth control pill use and the risk of heart clots.

	Heart Clot	No Heart Clot
Pill Users	23	34
Non-Pill Users	35	132

Investigate whether there is an association between birth control pill use and the risk of heart clots. Use a significance level of $\alpha = 5\%$.

Kahoot!

(x5)

Example 2 – Continued

In a study (USA, 1975), the association between birth control pill use and the risk of heart clots was investigated.

	Heart Clot	No Heart Clot
Pill Users	23	34
Non-Pill Users	35	132

Estimates in each sample

$$\hat{p}_1 = \frac{23}{57} = 0.4035, \quad \hat{p}_2 = \frac{35}{167} = 0.2096$$

Pooled Estimate:

$$\hat{p} = \frac{23+35}{57+167} = \frac{58}{224} = 0.2589$$

Example 2

- Go to today's Python notebook in VS Code
  - "Example: Contraceptive pills and risk of blood clots"



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Hypothesis test for multiple proportions

Comparison of c proportions

In some cases, you may be interested in assessing whether two or more binomial distributions have the same parameter p, i.e., testing the null hypothesis:

$$H_0: \quad p_1 = p_2 = ... = p_c = p$$

against the alternative hypothesis that these proportions are not equal (i.e., at least one is different).

Hypothesis test for multiple proportions

Table of observed counts for c samples:

	Sample 1	Sample 2	...	Sample c	Total
Success	x_1	x_2	...	x_c	x
Failure	$n_1 - x_1$	$n_2 - x_2$	...	$n_c - x_c$	$n - x$
Total	n_1	n_2	...	n_c	n

Common (average) estimate:

Under the null hypothesis, the estimate for p is:

$$\hat{p} = \frac{x}{n}$$

Hypothesis test for multiple proportions

Common (average) estimate:

Under the null hypothesis, the estimate for p is:

$$\hat{p} = \frac{x}{n}$$

"Use" this common estimate in each group:

If the null hypothesis is true, we expect the jth group to have e_{1j} successes and e_{2j} failures, where

$$e_{1j} = n_j \cdot \hat{p} = \frac{n_j \cdot x}{n}$$

$$e_{2j} = n_j(1 - \hat{p}) = \frac{n_j \cdot (n - x)}{n}$$

Hypothesis test for multiple proportions

Table with the *expected* counts in the c samples:

e_{ij}	Sample 1	Sample 2	...	Sample c	Total
Success	e_{11}	e_{12}	...	e_{1c}	x
Failure	e_{21}	e_{22}	...	e_{2c}	$n-x$
Total	n_1	n_2	...	n_c	n

General formula for calculating expected values in contingency tables:

$$e_{ij} = \frac{(\text{Row total } i) \cdot (\text{Column total } j)}{\text{total}}$$

Calculation of the test statistic - Method 7.20

The test statistic is

$$\chi^2_{\text{obs}} = \sum_{i=1}^{2} \sum_{j=1}^{c} \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

where o_{ij} is the *observed* count in cell (i, j) and e_{ij} is the *expected* count in cell (i, j).

Find p-value or use critical value – Method 7.20

Sampling distribution of the test statistic (under H_0):

χ^2 distribution with $(c-1)$ degrees of freedom (approximate)

Method with critical values:

If $\chi^2_{\text{obs}} > \chi^2_{1-\alpha}(c-1)$, then reject the null hypothesis.

Rule of thumb for validity of the test:

All expected values $e_{ij} \geq 5$.

Example 2 – continued

The *observed* values o_{ij}

Observed	Blood clot	No blood clot
Pills	23	34
No pills	35	132

Example 2 – continued

Calculate the *expected* values e_{ij}

Expected	Blood clot	No blood clot	Total
Pills			57
No pills			167
Total	58	166	224

Example 2 – continued

Use the "rule" for expected values four times, i.e.:

$$e_{22} = \frac{167 \cdot 166}{224} = 123.76$$

The *expected* values e_{ij}:

Expected	Blood clot	No blood clot	Total
Pills	14.76	42.24	57
No pills	43.24	123.76	167
Total	58	166	224

Example 2 – continued

Test statistic (include all cells):

$$\chi^2_{\text{obs}} = \frac{(23-14.76)^2}{14.76} + \frac{(34-42.24)^2}{42.24} + \frac{(35-43.24)^2}{43.24} + \frac{(132-123.76)^2}{123.76}$$
$$= 8.33$$

The critical value:

$\chi^2_{1-\alpha}(c-1)$ for $\alpha = 0.05$ and $c = 2$ (2 samples): 3.841

Conclusion:

Since $\chi^2_{\text{obs}} = 8.33 > 3.841$, reject the null hypothesis.

Example 2

- Go to today's Python notebook in VS Code
  - "Example: Contraceptive pills with χ^2"



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Example 3: Analysis of a contingency table

A 3×3 table: 3 samples with 3 categorical outcomes

	4 weeks	2 weeks	1 week
Candidate I	79	91	93
Candidate II	84	66	60
Undecided	37	43	47
	$n_1 = 200$	$n_2 = 200$	$n_3 = 200$

Is the voting distribution the same?

$$H_0: \quad p_{i1} = p_{i2} = p_{i3}, \quad i = 1, 2, 3.$$

Another type of contingency table

A 3×3 table: 1 sample with two variables with 3 categorical outcomes:

	bad	average	good
bad	23	60	29
average	28	79	60
good	9	49	63

Is there independence between the classification criteria?

$$H_0: \ p_{ij} = p_{i\cdot} p_{\cdot j}$$

Test statistic – regardless of table type: Method 7.22

In a contingency table with r rows and c columns, the test statistic is:

$$\chi^2_{\text{obs}} = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

where o_{ij} is the observed count in cell (i, j), and e_{ij} is the *expected count* in cell (i, j) (under the null hypothesis).

General formula for calculating expected values in contingency tables:

$$e_{ij} = \frac{(\text{Row total } i) \cdot (\text{Column total } j)}{\text{total}}$$

Find p-value or use the critical value - Method 7.22

Sampling distribution for the test statistic under H_0:

χ^2-distribution with $(r-1)(c-1)$ degrees of freedom.

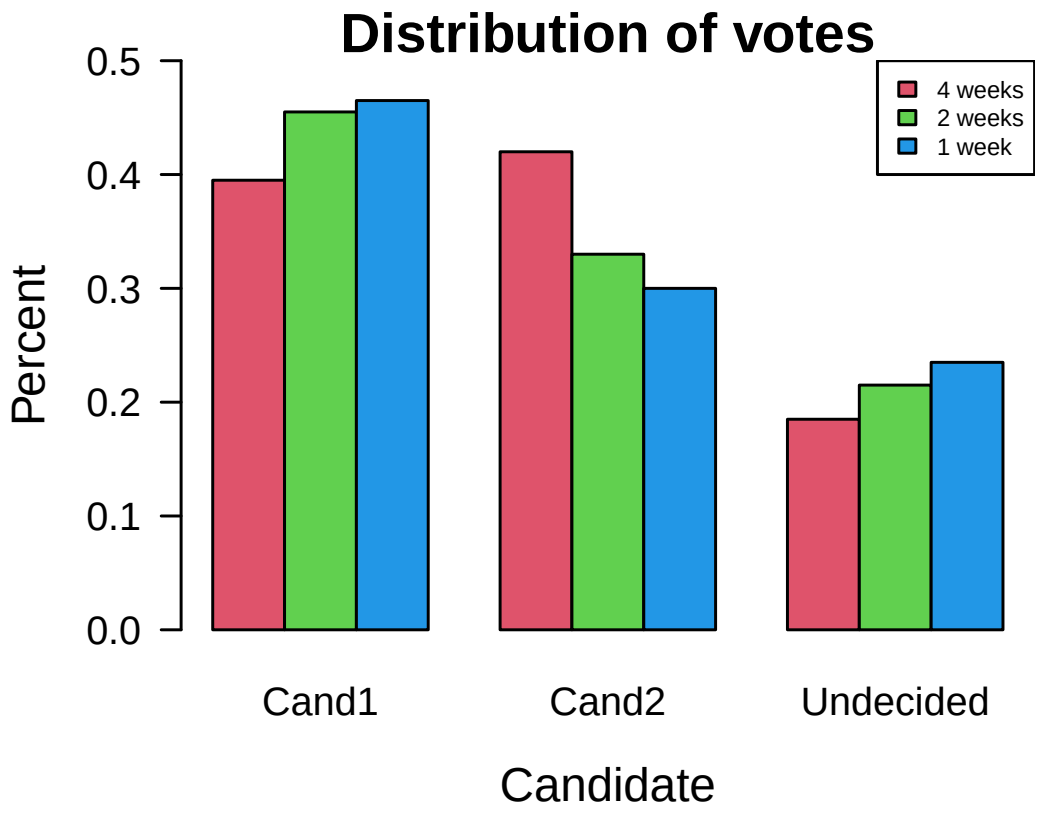
Method with the critical value:

If $\chi^2_{\text{obs}} > \chi^2_{1-\alpha}$ with $(r-1)(c-1)$ degrees of freedom, then reject the null hypothesis.

Rule of thumb for validity of the test:

All expected values $e_{ij} \geq 5$.

Example 3 – continued



Does the distribution change "significantly" over time?

Example 3

- Go to today's Python notebook in VS Code
  - "Example: Candidate votes over time"



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Today: Proportions (Proportions)

When the outcome/interest variable y_i is **binary** (yes/no, success/failure, 0/1)

- Proportion in a group: $\hat{p}$
- Relevant null hypothesis is often $p_0 = 0.50$ (not zero!)
- Comparison of proportions in two or more groups
- (Not included in this course: Proportion as a function of explanatory variable, logistic regression)

When the outcome/interest variable y_i is a category with > 2 groups

- Discrete distribution between the groups (one proportion in each group)
- Comparison of distribution, e.g., over time or for different "exposures".

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